

Designing for Dream Awareness through Somaesthetic Interaction

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Abstract

This thesis explores the experiential and reflective continuity between waking consciousness and the dreaming state through the lens of Research through Design (RtD) and somaesthetic interaction design. While commercial sleep technologies primarily focus on diagnostic surveillance and retrospective data quantification, this research addresses the "recall gap"—the rapid fragmentation and decay of dream memory upon awakening—by treating the sleeping body as an active channel for mindful somatic interaction.

Through the deployment of a *Dreamer's Pillowcase* cultural probe field study and a longitudinal, 24-hour autobiographical experiment ("24-Hour Reality Loop"), this project investigates how systematic, cross-state sensory cues combined with physical habit formation influence dream recall and nocturnal metacognition. The empirical findings reveal a distinct social and personal desire for dream agency, while isolating the "resleep" period as a highly receptive chronobiological window for sensory dream incorporation.

These insights informed the development of *The Metacognitive Bridge*, a wearable device concept that uses a localized, closed-loop software architecture to deliver non-invasive hybrid haptic-acoustic triggers. This thesis contributes to an alternative paradigm for wearable technology, shifting focus away from data extraction toward low-attention somatic scaffolding, proving that interactive artifacts can create reflective continuity across distinct states of human consciousness.

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1. Introduction

1.1 Problem Domain and Research Focus

This thesis explores how interaction design can help bridge the gap between our waking lives and our dreams. For decades, Interaction Design (IxD) has focused almost exclusively on making our waking hours more productive, leaving the "sleeping self"—a state of being that takes up nearly a third of our lives—largely ignored by designers (Hutchinson et al., 2003). But sleep is more than just a biological reset button. It is a rich psychological landscape where dreaming actively helps us handle stress, consolidate memories, and sort through our emotions, essentially functioning as a form of overnight therapy (Walker & van der Helm, 2009; Walker, 2017).

Even with these clear benefits, we usually feel completely disconnected from our dreams because our memories of them fragment and fade the moment we open our eyes (Stickgold, 2005). I call this friction the "recall gap." Current consumer sleep technologies do little to help; they focus almost entirely on passive tracking, reducing our rest to morning graphs and abstract metrics rather than encouraging any active, felt engagement (Li et al., 2015). I wanted to address this gap by designing an active, somaesthetic system that treats the sleeping body as a real participant in the interaction.

Grounded in Shusterman's (2008) somaesthetics and Höök's (2018) body-centric design principles, my research extends design attention directly into the dreaming state. Building on LaBerge's (1985) work with lucid dreaming—defined as achieving metacognitive awareness while asleep—this project does not treat lucidity as a rigid performance metric to guarantee. Instead, I explore how design can support the right physical conditions for mindfulness and self-reflection across both sleep and wake states. By testing how systematic sensory cues translate across the sleep-wake transition, I map out a practical loop for habit formation. My goal is not to deliver clinical or therapeutic outcomes, but to materialize a somatic design that helps users move away from passive dreaming and toward a reflective, continuous relationship with both halves of their lives.

1.2 State of the Art

To see where my project fits within modern technology, we can look at three main approaches: metric-driven consumer sleep trackers, emerging neurostimulation systems, and experimental design research dealing with dream interaction.

Commercial sleep tech has long favored numbers over experience. Popular consumer wearables like the Oura Ring Gen4 pack in sensors—tracking heart rate variability, movement, and blood oxygen levels—to map out our sleep stages. But the user experience here is strictly retrospective. You wake up to a breakdown of scores, percentages, and graphs detailing what happened hours ago. This abstract data ends up distancing us from the subjective, felt quality of our rest. Even screenless recovery bands like the Whoop 5.0, which try to cut down on daytime visual distractions, still tie us directly to smartphone apps for interpretation. They treat sleep as a post-event evaluation rather than something we can actively participate in while it happens.

On the other end of the spectrum is a new wave of "consciousness technology"—experimental neurostimulation setups built to directly alter how we dream. Devices like the Prophetic Halo try to identify REM sleep in real time using EEG and machine learning, then use transcranial focused ultrasound (tFUS) to stimulate parts of the brain responsible for executive reasoning. Similarly, setups like the Bía Smart Sleep Mask build electronics into a mask to guide lucid dreaming via light and sound cues. While these projects push technical boundaries, they bring massive physical and sensory friction into bed. Strapping rigid, complex hardware to your head conflicts with the soft, restorative environment needed for good sleep. These systems treat sleep as a problem to be optimized and engineered rather than a bodily experience to reflect on.

Closer to home, interaction design researchers have tried a more social approach. Rochegude's *Interactive Dreams* project at Malmö University looked at how a sleeper and an awake partner could interact through a Brain-Computer Interface (BCI). The system mapped the sleeper's emotional states onto a physical artifact that the awake partner could manipulate, sending tactile cues back to the sleeper through a specialized pair of pajamas. Rochegude proved that these sensory cues can weave cleanly into a dream narrative without waking the person up. However, the project also exposed a major flaw: multi-user dream setups create an uncomfortable power dynamic. The interaction quickly becomes about negotiation, vulnerability, and social performance between the two players, overshadowing private self-awareness.

Right now, sleep tech either tracks us passively, intervenes with heavy neurotech, or turns our dreams into a two-player game. Very few designers are looking at low-friction, wearable systems meant to help an individual build a continuous sensory habit across day and night. This is where *The Metacognitive Bridge* comes in. Instead of surveillance or external neural control, my project focuses on a standalone somatic design.

1.3 Framing the Design Territory

I chose to build the Metacognitive Bridge because of a massive, unmapped gap in sensory design, a recent maturity in DIY hardware, and a growing cultural frustration with consumer health tech.

The space between daytime mindfulness and nighttime dreaming is rarely explored in Interaction Design (IxD). Early projects in physiological computing, like *Brainball* (Hjelm, 2000), focused entirely on wakeful interaction, mapping a user's focus to move a physical ball in a game. Over the years, designers have built on these biological tracking ideas, but very few look at sleep. The closest design precedent is *DreamThrower* (Kamal et al., 2012), which used light and sound triggers during REM sleep to modify dream imagery and share those metrics online. Yet, even the few projects that tackle dream engineering tend to treat the sleeping body as either a passive data generator or a component in a video game. We are missing design research into self-contained, screenless physical systems built for individual habit formation across the sleep-wake boundary.

Studying dreams used to require high-friction laboratory equipment, where lucid dreamers had to look left and right in specific patterns (EOG signals) to send real-time Morse code messages to researchers outside (LaBerge, 2000). Today, the hardware landscape has completely changed. Small haptic actuators, open-source microcontrollers, and low-profile components let me bypass bulky lab gear entirely. This accessibility makes it possible to prototype a lightweight, localized wearable that safely taps into a user's sensory channels at home.

This exploration also addresses a broader cultural shift. Over the last decade, the Quantified Self movement has made tracking tools like Fitbit, Oura, and Whoop incredibly popular (Li et al., 2015). These devices brought sleep tracking into daily life and gave users a sense of digital agency over their nights. However, by reducing our rest to retroactive scores and medical-looking charts, these technologies detached us from the actual, felt experience of sleeping.

My project offers an alternative. It broadens what sleep technology can do by reframing the sleep-wake threshold as a rich site for self-reflection. I see this as a valuable path for interaction designers trying to work in low-attention spaces, and for people looking for screenless alternatives to data-heavy health trackers. The core purpose of this thesis is to design, make, and test a physical system that links waking reflection with narrative dreaming into a 24-hour cycle of personal awareness. By moving away from retrospective graphs and multi-user gameplay, I want to take sleep tech out of its clinical background and treat the dreaming mind as a valid space for personal interaction.

1.4 Research Question

> How can somaesthetic interaction design support reflective continuity between waking and dreaming states?

2. Theoretical Framework

2.1 The Neurobiology of Sleep and Dreaming

Sleep is not a passive blackout state, but a highly active biological process that keeps our bodies physically healthy and our minds sharp. Throughout the night, our brains move through repeating 90-minute blocks, shifting between Non-Rapid Eye Movement (NREM) and Rapid Eye Movement (REM) phases (Carskadon & Dement, 2011). Deep NREM sleep handles the physical heavy lifting—restoring cells, regulating metabolism, and using the glymphatic system to literally flush metabolic waste out of the brain tissue (Xie et al., 2013).

REM sleep is entirely different. During this phase, brain activity spikes and metabolic rates accelerate to near-waking levels, bringing on our most vivid, narrative dreams. To stop us from physically acting out these internal stories, the brain stem triggers a temporary state of full-body paralysis called REM atonia (Hobson & McCarley, 1977).

For an interaction designer, these unique physical boundaries are incredibly exciting. Because the body is paralyzed but the mind is highly active, external sounds or physical touches that stay just below the waking threshold can actually slip past the boundary of consciousness. The brain then weaves these real-world sensations directly into the dream itself without waking the person up (Stumbrys et al., 2012). This phenomenon creates a clear design territory for gentle, low-attention sensory loops that can operate across both our waking and sleeping hours.

Data also shows that remembering a dream depends heavily on what the brain is doing right before we open our eyes. Strong dream recall is tied to brief bursts of brain activity and tiny "micro-arousals" during sleep, especially in areas that process vision and senses (Scarpelli et al., 2021). These insights prove that our access to dreams is directly shaped by the shifting borders between sleep and wakefulness, which strongly supports my design's focus on precise timing, distinct physical cues, and low-friction routines to help pull those memories into the light.

2.2 Functional Theories of Dreaming

Dreams are essentially internal sensory and emotional simulations that actively help us sort through memories, process feelings, and practice real-world skills. During REM sleep, our brains drop their levels of stress chemicals like norepinephrine. This biochemical dip lets us safely replay emotionally intense memories without the usual panic or anxiety, acting like a form of overnight therapy that reorganizes our experiences (Walker & van der Helm, 2009).

From an evolutionary standpoint, theories like the Threat Simulation Theory suggest that dreaming works like a biological virtual reality headset, letting us rehearse how we would handle dangerous situations without any actual physical risk (Valli & Revonsuo, 2009). Other cognitive studies view dreaming as an "offline" sorting room where the brain takes our daily experiences, strips away the details, and integrates the core concepts into long-term memory structures (Stickgold et al., 2001).

The Continuity Hypothesis states that what we dream about is dynamically linked to our waking thoughts, dominant emotions, and recent life events (Scarpelli et al., 2021). The things we stress over, the conversations we leave unfinished, and even mundane daily tasks routinely show up in our dreams, proving that waking life and sleeping life are part of a continuous psychological thread. This cross-state bleeding follows predictable timing patterns. Boring, routine day-to-day events usually show up in our dreams within a day or two, while deeply emotional events might take several days to surface as the brain works to integrate them (Scarpelli et al., 2021). If our continuous daytime habits, thoughts, and moods systematically color our dream landscapes, then practicing a deliberate, physical sensory routine during the day should eventually show up in our sleep. This direct structural connection is exactly why I built the 24-hour interaction loop used in my self-experiment.

2.3 Neurophysiology

Brain scans show that when we enter REM sleep, our visual and emotional processing centers—like the amygdala and the parahippocampal cortex—light up with activity (Maquet et al., 1996). This neurological spike is exactly what gives dreams their hyper-vivid, emotionally intense, and deeply immersive textures.

On the other side, the dorsolateral prefrontal cortex (DLPFC) essentially goes dark (Muzur et al., 2002). Because this specific area handles logic, critical reasoning, working memory, and executive control, its deactivation explains why we passively accept completely bizarre, unstable, and impossible dream events without a second thought.

For an interaction designer, this physiological shutdown changes everything. Because a sleeper's capacity for critical, reflective reasoning is severely weakened mid-dream, I cannot expect a user to

suddenly initiate a complex, analytical thought process out of nowhere while asleep. Instead, achieving dream awareness has to rely on automated bodily habits, learned physical routines, and repeated sensory associations that are deeply trained during our waking hours. We have to prime the body while awake so the habit executes itself automatically when it encounters the same trigger while asleep.

Data also suggests that people who naturally remember their dreams easily show higher baseline activity in frontopolar brain regions (Scarpelli et al., 2021). These are the same areas we use for internally directed thought, autobiographical memory, and daytime mind-wandering, pointing to a direct link between a reflective waking mind and nighttime dream accessibility. Researchers using machine learning and neuroimaging have even managed to predict broad categories of dream imagery simply by looking at early sleeping brain patterns (Horikawa et al., 2013). While these studies are still exploratory, they confirm that dreaming uses the exact same neural machinery as waking perception, memory, and imagination. Instead of treating dreams as isolated nighttime glitches, interaction design can approach perception, memory, and dreaming as one continuous, accessible spectrum.

2.4 Metacognition and Lucid Dreaming

Lucid dreaming is the experience of waking up inside your own mind while remaining biologically asleep, realizing you are dreaming in the middle of a REM cycle. Brain scans show that when this happens, parts of the prefrontal cortex suddenly light up, allowing elements of self-reflection, logic, and executive control to come back online right inside the dream (Dresler et al., 2012).

Stephen LaBerge (1980) proved that we can systematically support this state using prospective memory techniques, like the Mnemonic Induction of Lucid Dreams (MILD) protocol. This method involves planting a clear intention while awake to recognize when you are asleep later that night. Similarly, practicing daily "reality tests"—where you regularly pause during the day to critically question if your environment is real—makes you far more likely to perform the exact same check inside a dream (Levitan & LaBerge, 1989). Brain imaging studies even suggest that frequent, natural lucid dreamers actually have more gray matter in frontopolar regions, which are the exact brain areas we use for introspection, self-monitoring, and deep reflection (Filevich et al., 2015). This means lucid dreaming is deeply tied to how reflective we are when we are wide awake.

It is much easier to become lucid if you have already trained your mind to remember your dreams in the first place. The more you recall your dreams, the more familiar you become with the unique layout of your own personal "dreamscape," making it obvious when you are navigating an impossible world (Noreika et al., 2010). This frames lucid awareness not as a random stroke of luck, but as a cognitive habit driven by repetition, focus, and physical routines. Instead of trying to force aggressive digital

control over the dream state using invasive tech, my project looks at how subtle, repeating sensory inputs can naturally bridge the gap between waking and sleeping mindsets.

2.5 Empirical Foundations of Sensory Incorporation

We usually think of sleep as a complete disconnect from the outside world, but a wealth of research proves that the sleeping brain keeps a window open to the environment (Salvesen et al., 2023). Instead of shocking us awake, sounds, touches, and smells that stay below our waking threshold are constantly monitored, filtered, and twisted by the nervous system to fit into whatever we are currently dreaming about. Scientists call this "dream incorporation" (Salvesen et al., 2023; Sammut & Conduit, 2012). For design researchers, this partial connectivity is incredibly exciting because it establishes a direct, non-invasive channel to communicate with the human subconscious (Salvesen et al., 2023).

Past experiments in dream engineering span several different senses, giving us clear rules for how specific environmental cues alter our dream worlds and emotional states:

> The Auditory Subsystem: Our hearing remains surprisingly resilient during REM sleep, which makes sound a highly reliable tool for dream modulation. Early lab work found that brief acoustic tones can slip into a dream either as random environmental background noise or disguised as part of the plot (Berger, 1963). Recent studies confirm that if we pair a specific sound with mindful daytime conditioning, replaying that sound during stable REM sleep acts as a mental spark, reactivating a critical mindset to induce lucid dreaming (Konkoly et al., 2024). When these gentle acoustic chimes or soft vibrations are timed precisely to REM cycles, the brain smoothly pulls them into the ongoing dream scenario without breaking the user’s sleep (Salvesen et al., 2023).
>
> The Tactile Subsystem: Touch is a direct, highly potent channel for physical sleep interaction. Classic experiments by Dement and Wolpert (1958) showed that spraying water droplets onto a sleeper’s exposed skin consistently altered their dream stories, forcing the brain to generate sudden scenes involving rainstorms, indoor leaks, or falling into water. Modern research confirms that tactile inputs completely bypass REM-related muscle paralysis ($atonia$). Localized haptic pulses and skin vibrations reliably break through the dream barrier, directly shaping how a user perceives their physical orientation and bodily agency inside the dream (Salvesen et al., 2023). Unlike multi-user systems that require an active, awake player to manually watch sleep stages and send triggers (Rochegude, 2015), my design relies on a self-contained, automated framework. Cutting out a second participant removes the ethical and psychological complications of an asymmetric power dynamic, letting the user manage their own consciousness tracking in total privacy.

The Olfactory Subsystem: Unlike sound or light prompts, smells rarely manifest as literal narrative objects—meaning you rarely see and smell a specific flower in a scene. Instead, research shows that scents color the global emotional architecture of a dream (Schredl et al., 2009). Introducing a positive aroma, like roses, systematically yields pleasant, happy dream reports. On the flip side, introducing a negative smell, like rotten eggs, triggers a sharp rise in nightmares and anxious dream moods.

The Visual Subsystem: Direct light stimulation during sleep usually relies on specialized sleep masks equipped with interior LEDs. When these light signals are pulsed during REM sleep, they penetrate closed eyelids and pop up inside the dream world as visual anomalies, showing up as flashing lightning bolts, pulsing emergency vehicle lights, or glowing celestial objects (LaBerge & Levitan, 1995).

These historical precedents prove that external sensory inputs do not have to disrupt our sleep. Instead, they can work as active structuring elements that systematically shape our dream cognition (Salvesen et al., 2023).

3. Methodology

3.1 Research through Design (RtD)

In this thesis, I use Research through Design (RtD) as my primary methodological framework (Zimmerman et al., 2010). RtD treats the actual process of making as a form of active inquiry, meaning that actionable knowledge is generated directly through building, iterating, and testing physical things. By developing tangible prototypes and exploring on-body interactions, I am investigating how physical sensations and everyday habits can shift awareness across the boundaries of waking and sleeping.

Instead of studying sleep through a clinical or diagnostic lens, this project focuses on the raw experiential and sensory qualities that surface when a person interacts with an artifact. I am using the design process itself as an exploratory, reflective tool to map out the fragile relationship between low-attention tech, bodily mindfulness, and the subjective reality of rest.

3.2 Field Study Strategy

For the field study, I deployed a "Domestic Probe" strategy built on the cultural probes methodology established by Gaver et al. (1999). Cultural probes rely on deliberate ambiguity, open-ended

questions, and playful prompts to capture deeply personal data from participants while they are in their own familiar environments.

Rather than chasing rigid, numerical data or charts, this approach gathers rich, layered text and sketches. These visual and written responses reveal internal values, emotional ties, and personal mindsets toward dreaming that would be impossible to catch using standardized surveys or structured interviews. This method is uniquely suited for digging into an intimate, fleeting experience like sleep, where clinical observation often fails to capture the true human texture of the night.

3.3 Autobiographical Design Strategy

To complement the collective field study, I used an autobiographical design approach to investigate the deeply personal, internal experience of the sleep–wake transition (Neustaedter & Sengers, 2012). I acted as the primary participant within a structured "24-Hour Reality Loop" experiment, documenting my own long-term responses to repeated sensory prompting and habit formation.

Within sleep and dream research, autobiographical design is incredibly valuable. Experiences like dream recall, lucid awareness, and the sensory distortions of transitional sleep are highly subjective. They are almost impossible to capture through detached, short-term user tests with strangers. By documenting my immediate reactions to haptic and auditory cues over an extended period, I was able to observe firsthand how these sensory interactions translated across both waking and sleeping contexts.

This approach also let me calibrate the physical interaction system on the fly. I systematically tweaked the timing, intensity, placement, and comfort of the sensory triggers based on my daily experiences. Material failures, sleep disruptions, and half-remembered dream fragments were given the same analytical weight as my fully successful lucid dreams.

Of course, autobiographical design has built-in limitations—mainly subjectivity, personal bias, and the risk of tuning a design too closely to my own eccentricities. To keep myself grounded, I integrated a long-term historical baseline into the study before starting the active intervention phase. I have kept a detailed, continuous dream journal between the ages of 18 and 25, mapping out my recurring dream landscapes, spontaneous lucid episodes, and baseline sleep habits. This eight-year archive established a clear comparative baseline for my natural dream recall and lucid frequency before introducing any hardware.

By comparing the findings of the "24-Hour Reality Loop" against years of my own pre-existing records, I was able to look at my findings in a broader longitudinal context. This helped me separate

my permanent, idiosyncratic sleep patterns from the experiences directly sparked by the new interaction design.

3.4 Ethical Considerations

Any research that digs into intimate matters such as personal dreams requires strict ethical boundaries. I structured this study from day one to align with standard academic protocols and the European Union’s General Data Protection Regulation (GDPR) rules for handling personal data.

Taking part in the *Dreamer's Pillowcase* cultural probe study was completely voluntary. Every participant signed an explicit, written informed consent form before I handed out any physical materials. They kept absolute ownership over their information, including the right to pull out of the project at any point without any penalty. To protect everyone's privacy under GDPR, I completely anonymized all written answers, sketches, dream diaries, and voice logs the moment I collected them. I assigned alphanumeric codes to the physical artifacts and stripped away all explicit personal markers—like names, specific nationalities, or exact locations—from my working database to make sure no one could be accidentally identified.

My longitudinal autobiographical experiment also put physical well-being, sleep quality, and safety first. During the early exploration phases, I carefully calibrated the vibration intensity, duration, and timing of the signals. I wanted to make sure the cues stayed gentle enough to avoid breaking my sleep cycles repeatedly or causing lingering daytime fatigue. By treating dream architecture as a private space for self-reflection rather than a target for data extraction or optimization, my design stops short of intrusive surveillance. I stored all my quantitative and qualitative logs on localized, offline storage drives to prevent network leaks or unauthorized access.

4. Field Study

4.1 Description and Rationale

To explore the elusive, volatile nature of human dreaming, this project deployed a qualitative cultural probe grounded in the foundational design-research framework established by Gaver, Dunne, and Pacenti (1999). The primary artifact developed for this field study was an unembellished, completely blank white pillowcase. This deliberate selection is rooted in somaesthetic design theory, which posits the physical, lived body as an active, central site for sensory experience, self-monitoring, and deep critical reflection (Shusterman, 2008; Höök, 2018). By introducing a probe directly into the

participants' personal sleep environments, the research intervention engages them at the immediate physical boundary between waking life and the subconscious.

The pillowcase functions here as a classic boundary object—a familiar, non-threatening bedroom textile recontextualized into an active canvas for qualitative reflection. In sharp contrast to rigid, clinical mobile applications or intrusive digital tracking tools, this soft, analog format drastically lowers the barrier to participation, reduces performance anxiety, and invites a raw, authentic, and highly personal form of expression. This material strategy allowed the inquiry to investigate how self-awareness navigates the threshold between waking and dreaming by embedding the very act of self-documentation directly into the physical context of sleep.

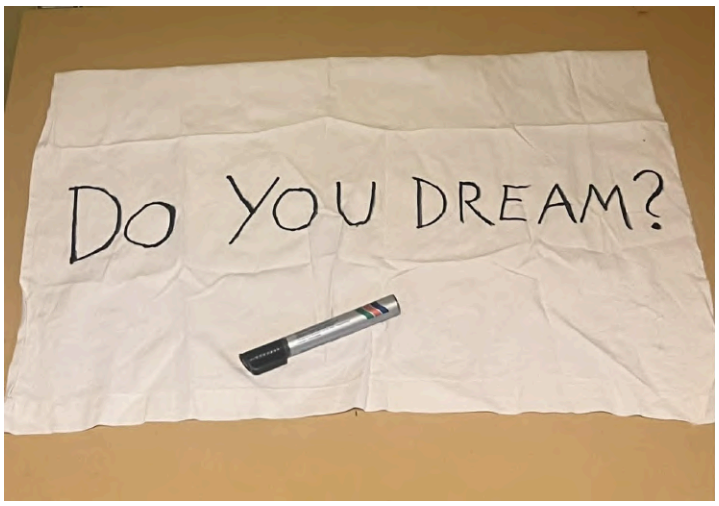


Figure 1: Cultural probe pillowcase.

4.2 Implementation

The field study was conducted with a group of twelve young adults from diverse international backgrounds during an intensive Erasmus+ youth exchange. This demographic provided a unique cross-cultural perspective on sleep behavior and dream mythologies. After introducing the participants to the wider theoretical scope of the research, they were provided with the unembellished white pillowcase and a set of black permanent textile markers.

To explore different temporal, cognitive, and imaginative dimensions of the dreaming self, two distinct prompts were hand-rendered directly onto the fabric surface:

> The Exterior Prompt ("Do you dream?"): This prompt was written boldly and visibly across the outside of the fabric canvas. It was engineered to capture immediate, everyday retrospective experiences of dream recall, effectively mapping out the user's historical and present relationship with their sleep architecture.

The Concealed Internal Prompt: Hidden deliberately inside the internal fold of the pillowcase, this prompt asked: *"If you were in a dream right now, what would you do?"* This question purposely shifted the participant's focus away from passive tracking to tap into their latent desires for agency, control, and spontaneous lucidity while asleep, effectively mapping out their future aspirations and imaginative goals.

Participants were explicitly encouraged to write, sketch, annotate, or doodle completely freely over the course of the study. The onboarding process was intentionally kept informal, open-ended, and playful. This relaxed setting was vital for establishing a safe psychological space that supported uninhibited, authentic self-expression and bypassed the clinical detachment often caused by traditional sleep-tracking technologies.

4.3 Material Analysis

Following the deployment phase, the physical pillowcases came back completely filled with an uninhibited mix of handwriting, text fragments, and spontaneous sketches. By the end of the study, the blank fabric had transformed into an expressive, overlapping canvas of collective human experience. Through a systematic qualitative coding process, I classified these hand-rendered answers into four broad directions:

Direct Affirmation or Negation: Bold, definitive statements like *"Yes"*, *"YES"*, *"No"*, and *"Nope"*.

Uncertainty and Inconsistency: Shaky, faint annotations capturing the volatile nature of memory decay, including *"Sometimes"*, *"I don't know"*, and *"Can't remember"*.

Aspirational Longing: Expressive notes indicating a deep desire to access sleep memories, typified by the repeated phrase *"I wish"*.

Spontaneous Doodles: Playful illustrative fragments—including smiley faces, hearts, and classic comic representations of sleep (*"zzzzz"*). These gave the textiles a deeply human tone and proved that the participants felt a strong emotional connection to the probe.

The physical annotations were not written in neat, uniform lines or structured grids; they were fragmented, layered, and turned at erratic angles all over the fabric surface. This visual messiness serves as a direct material mirror of how varied, inconsistent, and deeply personal our relationships with our dreaming minds actually are.

A significant portion of the written data shines a light directly on this volatility. The high concentration of phrases like *"sometimes," "I don't know,"* and *"can't remember"* suggests that the

primary breakdown in dream interaction is not an absence of dreaming itself, but a complete lack of structural support and sensory continuity right at the sleep-wake threshold. Writing down *"I wish"* exposes a disconnect between what a user actually experiences every morning and the creative relationship they want to achieve with their subconscious.

I was able to further contextualize these material findings during subsequent group discussions. Participants described dream memories as something incredibly fragile—a faint sensory trace that vanishes almost instantly upon waking unless they actively talk about them, write them down, or emotionally connect with them within the first few minutes of wakefulness. Interestingly, several individuals noted that their dream recall became dramatically stronger when they talked about their dreams out loud during informal morning routines, such as breakfast conversations.

This collective insight strongly points to the existence of a *Recall Gap*: a sharp drop-off between the hyper-vivid dream experience itself and our biological ability to consciously hold onto it when we wake up. Without immediate interactional support, repetition, and reflection in daily life, dream memories rapidly decay.

Even though everyday dream recall was experienced as incredibly fragile, the underlying desire to exercise active volition inside those dreams was remarkably strong across the entire group. The answers to the hidden internal prompt revealed deep imaginative cravings for freedom, exploration, and creative control over the subconscious environment.

By far, the most universal wish recorded inside the pillowcase folds was the desire to fly. What struck me during the analysis was that while almost every participant understood the concept of lucid dreaming, only a tiny fraction actually experienced it on a regular basis. This highlighted a massive disconnect between knowing that lucid dreaming is possible and actually possessing the behavioral tools to wake up mentally inside a dream plot.

This contrast between unreliable dream access and a strong desire for control is highly significant. It proves that users are not indifferent to their sleep experiences; they possess an unmet desire to engage with their subconscious, which gave this project a clear mandate to design a hardware system capable of fulfilling that need.

When evaluating what a dream actually feels like during the group discussions, participants made it clear that dreams are not isolated, clinical nighttime anomalies. They are emotionally intense events that consistently spill over into waking life. Many people described how a dream's mood can linger throughout the following day—shaping their morning interactions, dictating their energy levels, and leaving behind a distinct emotional texture that alters their wakeful perception. Waking life and sleeping life do not exist as separate categories; they form a continuous, unbroken emotional loop.

When describing the sensory qualities of the dream state, participants noted that they experience them from an immersive, first-person perspective. Physical feelings—especially the sensation of touch—were incredibly common, which tells us that the dreaming self retains a strong, active sense of bodily awareness while asleep.

Almost everyone had stories about real-world sounds, noises, or touches bleeding directly into their dream plots. Morning alarms, ambient background music, or a partner moving in bed routinely weave themselves into the narrative logic of the dream instead of snapping the person awake. For instance, a rhythmic car alarm outside might manifest inside the dream as a drum beat or a warning siren within the story.

Participants noted that this blending happens most often during morning "snooze" or "resleep" periods—those moments where you wake up briefly but drift right back to sleep. They characterized dreams occurring during this transitional window as far more vivid, coherent, emotional, and easy to remember than dreams from earlier in the night.

This provides proof that subtle, designed sensory inputs can enter a dream without ruining sleep quality or waking the sleeper. The late-morning sleep-wake boundary stands out as the perfect chronobiological window for a sensory intervention.

4.4 Emerging Design Implications

The combined results of the *Dreamer's Pillowcase* cultural probe and the follow-up interviews expose a clear, systemic tension between the participants' everyday, unassisted sleep experiences and their personal goals for self-awareness. While routine dream recall was consistently described as fragile, fleeting, and highly erratic, the underlying desire for personal volition, control, and deeper engagement with the dream state remained remarkably strong across the user group.

This exact tension was visibly written all over the annotated surfaces of the pillowcases, where responses constantly fluctuated between a stark *"No"* and an emphatic *"Yes"*, almost always mediated by explicit notes of hesitation or uncertainty like *“sometimes,” “I don’t know,”* or *“can’t remember.”* Taken together, these findings prove that the core problem is not that people aren't dreaming, but rather that they lack structured access and sensory continuity across the sleep-wake threshold.

The real tactical goal of this project therefore becomes structurally supporting a user's reflective awareness of the dream narratives that are already happening naturally. The primary design challenge is helping users recognize, access, and hold onto dream memories across the slippery sleep-wake boundary, minimizing the effects of the *Recall Gap*.

This realization is what establishes *The Metacognitive Bridge* as the central conceptual direction of this thesis. Rather than functioning as a clinical diagnostic tracker, the system is positioned as a reflective interaction framework. It is designed to strengthen the physical continuity between waking awareness and dream awareness through repeated sensory habits and everyday embodied routines.

The qualitative success of the hidden internal prompt proves that interaction design for low-attention states needs to look far beyond clinical quantification, numeric sleep scores, or purely biometric feedback. Participants responded much more strongly to playful, ritualistic, and emotionally meaningful forms of interaction than to abstract, data-heavy tracking systems that leave them feeling cold.

As a result, the design direction prioritizes subtle, embodied interactions that fit naturally into a person's existing bedroom routines. The system emphasizes low-attention sensory feedback, tactile body loops, and reflective habit formation as the primary tools for supporting self-awareness across transitional states of the mind.

5. Autobiographical Experiment

5.1 Description and Rationale

The autobiographical experiment functions as a continuous, 24-hour closed-loop interaction framework. Serving as the primary research subject, I adopted a first-person designer-researcher strategy (Neustaedter & Sengers, 2012). The study combined a custom hardware routine with prospective memory priming, split into two distinct phases:

A mobile device was programmed to deliver unique, highly specific haptic vibration profiles at completely random intervals throughout the day. The moment I felt this pulse against my arm, I had to snap out of my daily cognitive autopilot, anchor my focus within my immediate body, and perform a rigorous "Reality Check." These audits involved counting my fingers, checking if digital clock faces or text remained stable when I looked away and back, and testing physical boundaries by trying to push my palm straight through solid walls or desks.

Throughout the night, the exact same hardware setup remained strapped to the identical position on my arm. I scheduled the system to trigger those same rhythmic haptic pulses—this time paired with a subtle acoustic chime—during my dense, late-morning REM cycles. The goal was sensory incorporation: letting the cue slide smoothly into my ongoing dream narrative without shocking my body awake. I meticulously documented both phases in a structured design journal right before falling asleep and immediately upon waking.

Using an autobiographical approach alongside my eight-year personal dream archive (spanning ages 18 to 25) was crucial for a few key reasons:

Deeply internal experiences like dream recall, lucidity, and sensory processing during transitional sleep are incredibly fragile and volatile. It is logistically and ethically unfeasible to capture these fleeting cognitive shifts through detached, one-time user tests with outside participants. Using my own lived body as the primary design material allowed me to calibrate trigger intensities, optimize physical comfort, and analyze dream fragments in real time against a granular, lifelong understanding of my own sleep baselines.

For a sensory input to act as a clear cross-state catalyst, it has to maintain strict material and spatial consistency. If the nighttime prompt differs even slightly in rhythm, intensity, or body location from the waking cue, it can produce a totally different outcome. Continuous self-testing allowed me to standardize wearing the hardware on my wrist across the full 24-hour loop, successfully training the necessary neuromuscular habit.

5.2 Pre-Experimental Content

Before starting my design intervention, I ran a systematic qualitative analysis on my own longitudinal dream archive, which covers eight years of continuous entries from age 18 to 25. In autobiographical design research, a personal archive like this is a massive empirical asset—it allows me to treat my own lived body and historical subconscious as primary design materials (Neustaedter & Sengers, 2012). By breaking down the structural, environmental, and mental patterns of my natural, unassisted lucid dreams over a multi-year period, I could move past generic assumptions and design directly for the actual limits of my own dreamscape. Looking back through the entries, I could see what tended to spark my lucid awareness.

1.

(I tried repeatedly to wake up from my ex-partner's sofa, genuinely believing I was awakening each time, though I simply kept dreaming). I "woke up" on the sofa in the TV room at my grandfather's house. One of my sisters was sleeping there as well, and my grandfather was sitting in front of me, talking through all the mechanics assignments I was supposed to finish. I felt a deep sense of stress.

Afterward, I went down to the kitchen, and my grandfather was there too. We discussed my schoolwork, and he told me, *"I've read all of yours and your sister's English assignments, and hers is 10% better than yours."* I replied that it didn't really matter, as long as one's reading comprehension

was good. Then, strangely, he started play-boxing with me. I defended myself half-heartedly, but it felt weird and uncomfortable.

I then carried a container filled with eyes into the living room and began eating out of it. They looked exactly like human eyes, yet inside the dream, it felt completely logical that they tasted delicious. After a moment, a deeper layer of reflection kicked in, and I thought to myself: *Isn't it actually pretty disgusting to eat eyes?* I understood that I was dreaming. Immediately after that, I flew! I spoke aloud, and I could hear my own voice with terrifying, crisp clarity; it even echoed through the space.

2.

(I thought I woke up, kicked out physically, and spoke out loud in my actual sleep). I was able to read the screen of my mobile phone, and the clock face read 13:00 inside the dream. I was on a sofa belonging to my ex-partner. I sat up and called out my ex-partner's name to see if he was awake. I walked over and poked him to wake him up.

Looking up, I noticed a heavily blurred window high up in the ceiling, with anonymous people walking past outside. I tried to orient myself by finding another wall and walked into a bathroom lined with blue tiles. An unfamiliar blonde girl walked in and asked if we could help her measure a doorway. My ex-partner and I were wrapped up in blankets. I asked her, *"Do you have a bevel square?"* She said yes, and I asked, *"Where do you live?"* She replied, *"Just over here."* The scene abruptly skittered into darkness, transforming into beach houses packed with people.

3.

(I tried repeatedly to wake up in my own bed at home after attending a university lecture. I experienced intense, heavy sensations of floating, completely unable to determine whether I was actually awake or still asleep). I "woke up" inside a false awakening. One of my sisters walked into the room, looking and acting strange. To test if I was dreaming, I pinched my own skin to see if it would hurt. I then exposed myself to my sister to see if she would react, but she remained entirely indifferent. Desperate for a definitive answer if I was dreaming, I intentionally dove headfirst toward the floor to see if it would hurt.

4.

(I half-slept and half-woke up many times in a row). I found myself right outside my house, possessing full control over my movements. I checked the mailbox, sorted through the post, and felt an impulse to walk over to a completely random house nearby. Suddenly, a systemic shift occurred: Everything went partially black, then I was launched into a lucid dream.

I could hear the distinct sound of footsteps walking over asphalt with incredible clarity. I saw my family members from behind, carrying bags and luggage toward the car as if they were setting off on a trip. I called out, *"Where are you guys going?"* My father replied, and I heard his voice so clearly that I consciously marveled, *Wow, it's incredible that my brain can perfectly recreate his exact voice like that.* I felt completely present, living entirely "in the moment." I attempted to step forward to follow them, but as I did, I felt my sleeping legs twitching in the real bed, and I woke up.

5.

(I don't remember how this one started). I was in a lucid dream where my family was preparing a dinner party at my grandparent's house. I consciously reflected that it didn't matter at all if I completely ignored the gathering, reasoning: *This is a dream, and they won't feel hurt or sad in the real world.*

I walked away and went into my grandfather's bathroom to look inside a storage cabinet, while a song played on a continuous loop in the background. I tried to use my mind to materialize a certain person in the kitchen, but his form failed to maintain structural stability—he appeared only as a faint, barely visible silhouette before the entire dream dissolved.

6.

(I kept falling asleep and waking up over and over again, perceiving every single sensation with intense, hyper-vivid clarity while lucid dreaming). In the dream I was at a chocolate buffet. A guy I liked was standing there, holding a bowl of chocolate spread. Realizing I possessed full agency and control, I resolved to push the boundaries of the scene: *Now I am going to do something.* I stepped forward, dipped my finger into the spread, took hold of the back of his neck, and smeared it across his cheek before licking it off. I then applied some to my own cheek, directing him to lick it off in return, which he did.

7.

I discovered rabbit droppings scattered across my bedroom floor. I searched the space and found a small, grey-and-white rabbit. When I picked it up, it began squirming wildly in my hands and aggressively trying to bite me as I walked. The way it moved and felt in my hands was very unnatural. I thought, *"This is a dream."*

I walked outside and encountered a Volkswagen van packed with guys. Bypassing all social hesitation, I walked straight up to one, climbed inside, and initiated contact. The scene transitioned to a school courtyard filled with people.

I decided to test my creative intent further, thinking: *It would be great if this guy was here right now.* After a brief delay, a high doorway swung open and he walked in behind a group of other people. He was wearing a sharp suit with messy hair that was being blown about because it was windy. He was trying to fix it with his hand as he walked. We stepped toward each other, and I remarked, *"Oh, look at your hair."* I then licked on my hand and smoothed his hair into place.

8.

I was sitting at a table with two of my sisters. I tried to use my mobile phone to look at a map interface, but discovered that the written text was highly distorted, strange, and completely illegible. This typographic failure triggered an immediate thought: *Am I dreaming?* I leaped onto the table and confronted one of my sisters, asking what the time was. She responded with a complete breakdown of conversational logic, stating: *"A very nice place to be."* I rejected the nonsensical reply, demanding: *"No, what is the exact time? I am late."* She muttered another strange phrase. I turned to my other sister, asking explicitly if I was locked inside a dream. I then asked them both to help me fly by throwing me into the air, which caused the landscape to lose visual fidelity and dissolve.

9.

I was in a lucid dream and tried to speak to one of my sisters, explicitly announcing to her that we were inside a dream state. The moment I made this declaration, the entire environment suffered a catastrophic collapse, turning completely black as I experienced a terrifying sensation of falling through an empty void.

The loop transitioned into awakening in my actual bed. However, I discovered I was suffering from physiological sleep paralysis: I possessed full conscious awareness of my physical body and bedroom environment but was completely unable to initiate any movement. I was trapped in a state of bodily immobility.

10.

I saw myself in the mirror in the bathroom. The skin above my upper lip was black. I thought it was strange. I realized I was dreaming. Euphoria, it got dark and I flew away. But then I woke up.

5.2 Dream Database Analysis

To translate these archival logs into actionable requirements for my project, I analyzed the ten entries using thematic qualitative coding. Instead of looking at the psychological or symbolic meaning of the

dream images, I focused entirely on the structural, sensory, and cognitive friction points—essentially mapping out the physical, temporal, and logical boundaries of my unassisted mind.

I broke this analysis down into three core dimensions: the timing and sensory cues that naturally sparked my dream awareness, the emotional and creative triggers that accidentally broke and dissolved the dreams, and the cross-state physical qualities I can leverage for interaction design.

Triggers of Realization

Looking closely at the archive, I noticed that self-awareness rarely emerges from abstract, top-down logical deduction while asleep. Instead, my lucidity was almost always sparked by two distinct factors: a specific morning sleep window or a direct collision with a physical aberration.

> A massive percentage of my successful lucid dreams followed a distinct temporal pattern of fragmented sleep-wake transitions. As I documented in Logs 1, 2, 3, 4, and 6, I frequently experienced states of *"half-sleeping and half-waking many times in a row"* or repetitive false awakenings where I genuinely believed I had woken up on a couch or in my bed. This natural sleep fragmentation isolates the late-morning transition phase as a highly receptive window. It is a moment where brain activation rises enough for critical self-reflection to come back online, even while the body remains paralyzed under REM atonia.
>
> When my lucidity occurred outside of this groggy morning loop, it was usually driven by an impossible dream physics or anatomy. In Log 1, lucidity was sparked by a sudden behavioral contradiction (*"Isn't it actually pretty disgusting to eat eyes? I understood that I was dreaming"*). In Log 7, the prompt was a strange, tactile distortion (*"The way it moved and felt in my hands was very unnatural"*), while in Log 10, it was a sudden visual error encountered while looking in a bathroom mirror (*"The skin above my upper lip was black. I realized I was dreaming"*).

Triggers of Dissolution

Sustaining awareness inside a dream is incredibly difficult. The archive exposed a direct link between trying to exert too much control, or experiencing intense emotions, and the immediate collapse of the dream environment.

The exact moment I realized I was dreaming, the sudden rush of consciousness often triggered a massive emotional spike. In Log 10, my realization was instantly followed by a wave of intense euphoria, which caused the room to rapidly darken before shocking me wide awake. This emotional excitement overstimulates the nervous system, forcing the brain out of the stable REM state required to hold the dream together.

Trying to forcefully manipulate the environment or manifest specific things routinely ruined the dream stability. In Log 5, I tried to use direct mental command to materialize a specific person in the kitchen. My dreaming brain failed to compute this top-down request, rendering only a faint, unstable silhouette before the entire scene dissolved into nothing.

Breaking the internal social or structural rules of a dream world often breaks the simulation itself. In Log 8, jumping onto a table to aggressively confront a character caused the landscape to lose its visual resolution and melt away. Even worse, explicitly telling a dream character that we were inside a dream in Log 9 triggered an immediate void crash—the entire environment collapsed into pitch blackness and left me falling through an empty void, dropping me straight into a state of sleep paralysis.

Cross-State Somaesthetic Frameworks

This baseline analysis highlighted an essential design paradox: while the dreaming mind completely struggles with language, digital interfaces, and text, its ability to simulate rich, complex physical feedback remains flawlessly intact.

In Log 2 and Log 8, my phone perfectly mirrored the physical weight and glass form factor of my real hardware, but in Log 8 the user interface suffered fatal rendering errors. The map display text was highly distorted and completely illegible. Conversational logic collapsed in the exact same way; when I demanded factual data like the exact time in Log 8, a character produced a grammatically correct but illogical sentence fragment: *"A very nice place to be."*

In direct contrast to this digital and linguistic collapse, my brain simulated physical textures, tastes, and acoustic details with perfect accuracy. This includes the exact taste of delicious food, the visceral feeling of chocolate spread on skin (Log 6), the crisp sound of footsteps walking over asphalt (Log 4), and my father's voice echoing so clearly that I remember marveling at how my brain could perfectly replicate his exact vocal timbre.

By shifting from a narrative diary to a structural analysis of these ten logs, I transformed my long-term records into a definitive set of design requirements for *The Metacognitive Bridge*:

Since my natural lucidity was so tightly linked to the morning transitional loop (Logs 1, 2, 3, 4, 6), the design should focus on that specific time. Because intense emotions (Log 10) or sudden, jarring mental shifts (Log 9) trigger void crashes and sleep paralysis, the nocturnal prompt must be incredibly gentle. It has to deliver a low-intensity, comforting sensory signature that slides smoothly into ongoing dream plots without startling the nervous system. The hyper-fidelity of my somatosensory channels—including detailed touch textures and auditory clarity—proves that physical sensation is our most reliable cross-state pathway.

5.3 Experimental Setup

The active testing phase of this project revolved around a continuous 24-hour cycle designed to see if a cross-state sensory bridge would actually work in practice. To build a continuous loop of physical awareness, I combined a custom hardware routine with classic psychological priming techniques. I split the daily experiment into two main phases:

Phase 1: Daytime Habit (Metacognitive Priming)

During the day, my focus was on building an automated awareness routine. I wanted to pair a distinct physical sensation with a sudden, intentional pause for self-reflection. To do this, I had to move away from standard smartphone alerts, which our brains instantly link to wakeful stressors like text messages, work emails, and calendar reminders. I completely muted all normal ringtones and notification sounds, and instead programmed a custom vibration profile using my mobile device's native haptic engine.

I designed the signal to use a specific, rhythmic pattern of short, deliberate pulses that felt completely different from a standard phone buzz. I configured these custom triggers to fire at completely randomized intervals throughout the day so that they remained unpredictable, forcing me to snap out of autopilot and redirect my attention to my body and surroundings.

The moment I felt this localized physical pulse, I paused whatever routine task I was doing to perform a conscious "Reality Check." Based on the glitch patterns identified in my archival analysis, I systematically questioned the physical stability of my immediate environment using three specific checks:

- Text and Digital Clock Audits: I looked at a digital clock or a line of text, turned my head to break my line of sight, and then looked back to see if the letters warped, melted, or changed—a classic failure point for dreams.

- Material Boundary Testing (Solid-Push): I forcefully attempted to push my hand straight through a nearby solid surface, like a wall or a desk, to verify that physical matter was actually resisting my touch.
- Anatomical Audits (Finger Counting): I visually counted the fingers on one hand while focusing deeply on the skin-on-skin friction of my hand, checking for any physical inconsistencies.

Phase 2: Nocturnal Anchor (Sensory Priming)

The nighttime phase focused on delivering that exact same sensory cue to my mind while I was in the middle of peak dreaming windows. I scheduled the custom alarm to trigger after I was fully asleep, strategically timing it to hit the longer, denser REM cycles that happen in the morning. During these blocks, the brain is highly receptive to weaving external stimuli into a dream narrative without triggering a full awakening. Right before falling asleep, I reinforced this setup with MILD protocols, mentally repeating the intention: *"The next time I dream, I will realize I am dreaming."*

To maintain total sensory consistency across the boundary of consciousness, I strictly standardized how I physically wore the hardware. If I was actively working or using the phone during the day, I held the device directly in my hand. During passive periods, or when I was moving around, I strapped the phone firmly against my arm using an adjustable sports wristband. This guaranteed that the physical sensation of the custom rhythmic vibration was transferred directly to the skin of my arm at all times.

I kept the device strapped to my arm in the exact same physical position while I slept throughout the night. By keeping the location on my body, the pressure of the wristband, and the texture of the vibration completely identical between day and night, I created the exact material conditions needed for a single somatic signal to work double-duty: functioning as a waking mindfulness prompt by day, and a subconscious dream anchor by night.



Figure 2: Experiment wearable during the day and night.

5.4 Execution

I kept a daily log within a custom Design Journal to capture both my numerical behavioral habits and my raw, physical reflections. I made a strict rule to fill out entries before sleeping and the very second I woke up. Writing instantly upon awakening was the only way to beat rapid memory decay and preserve the raw texture of my sleep experiences. The journal was structured around two distinct layers of data:

> Quantitative Parameters: I logged the exact times I fell asleep, my morning wake-up times, and the total duration of my sleep. I kept explicit binary tracking (Yes/No) of whether I actually noticed the haptic trigger while asleep, and whether it accidentally shocked me wide awake. I also documented whether I successfully achieved lucidity. If I did, I categorized the source of that awareness—tracing it back to either the long-term mental habit of my daytime training or the direct, immediate impact of the nighttime vibration.
>
> Qualitative Reflections: Beyond the numbers, the journal captured the nuanced, felt experience of this cross-state interaction. I reflected deeply on the physical intensity, the rhythm, and the bodily comfort of the wristband setup to help me figure out how to refine the final wearable enclosure. I also logged exactly how my brain interpreted the signal inside my dream plots—noting whether it was smoothly swallowed by the dream imagery or if it acted as a spark for sudden self-awareness. I tracked my shifting moods during the daytime priming

sessions, mapping out how the randomized prompts felt during the day. Combining these layers gave me deep insight into how the design was perceived across both halves of my day.

6. Empirical Findings

6.1 Results of the Autobiographical Experiment

6.1.1 The Waking Experience: Habit Formation

During the daytime phase, my device fired off custom triggers at completely randomized intervals. Because these signals were entirely unpredictable, they successfully broke through my daily autopilot routines and forced me into moments of heightened awareness. Each time my arm pulsed, I had to consciously pause whatever task I was working on and actively redirect my attention to my body and the immediate environment.

The reality checks quickly became a set of deeply familiar physical actions. I would stop, hold up my hands, and visually study their physical appearance, focusing intensely on the lines of my skin. I also closely checked nearby lines of text, digital clocks, or tiny environmental details to see if anything was warping or glitching, and I tested my physical boundaries by attempting to push my palm flat through solid desks or walls.

Repeating these exact checks dozens of times throughout the day completely changed how the interaction felt. It became a deeply embedded bodily habit.

Over time this routine began to offer unexpected benefits outside the scope of dream engineering. The randomized prompts started working as a valuable waking practice for mindfulness, offering a brief mental reset and a moment of grounding in the middle of a stressful workday. This suggests that localized, somatic prompting has genuine value for wakeful self-reflection.

6.1.2 Comparing Trigger Mechanics and Device Placement

Testing different vibration patterns, sound volumes, and device locations showed me that the physical layout of a signal completely dictates how your brain handles it while you sleep. Simply tweaking the duration, intensity, or rhythm changes whether a cue jolts you wide awake, goes unnoticed, or blends smoothly into an ongoing dream.

I found that auditory signals were by far the most reliable way to get the brain's attention during sleep. Soft tones and chimes frequently drifted into my dream experiences without shocking me awake,

proving that a quiet, low-intensity sound can easily sit right below the waking threshold while keeping a line of communication open to the mind.

Haptic signals were much more unpredictable. Sometimes a vibration pattern vanished completely into the night, leaving behind zero memory when I woke up. Other times, I would wake up with a crystal-clear sensory memory of feeling the physical pulse against my skin. This inconsistency showed me that on-body haptics depend heavily on how deeply you are sleeping and where you are in your sleep cycle.

This unpredictable boundary maps perfectly onto what prior design researchers found. Rochegude (2015) discovered that during early user tests of vibrating sleep garments, participants seamlessly swallowed intense, continuous physical buzzes into their dream plots—interpreting the physical vibration as the narrative sensation of getting a tattoo or hearing a buzzing restaurant food pager—all without waking up. Rochegude noted that standalone haptics hit a wall because sleepers rarely recognize the buzz as a real-world cue once they wake up. They just assume it was part of the dream story.

To fix this, I realized that the brain cannot be left to deal with a standalone tactile buzz. The vibration is much more effective when paired with a harmonized acoustic melody, creating a distinct, hybrid sensory prompt that is sharp enough to slice through automated dream logic.

Where I put the hardware also made or broke the entire interaction loop. Shoving the device inside my pillowcase was a complete failure; having a speaker and a vibration motor right against my skull caused abrupt, startled awakenings every single time. It completely ruined my sleep quality and made dream incorporation impossible.

Moving the device to my wrist was better. The skin on the arm experienced the vibration as a softer, less intrusive pulse, allowing the cue to echo more naturally through the sleep environment. Keeping the hardware on the wrist maintained absolute continuity with my daytime checks, reinforcing the physical link between my waking habits and my nighttime awareness.

6.1.3 The Chronobiological “Resleep” Window

An important discovery during my experiment was the direct link between my sleep cycles and the likelihood of actually waking up mentally inside a dream. I found that waking up briefly in the dead of night was great for standard dream recall, but it almost never led to a fully lucid experience. Instead, my highest probability of achieving lucid awareness happened during a late-morning transitional window that I call the “resleep” phase.

[Morning Awakening] ⟶ [Immediate Return to Sleep] ⟶ [Hyper-Vivid “Resleep” Dream]

Whenever I woke up briefly in the morning and immediately drifted right back to sleep, the resulting dreams became incredibly vivid, continuous, and emotionally intense. My chances of stepping into a lucid state skyrocketed during this exact window, which also perfectly backed up what the participants told me during my field study interviews.

This maps beautifully onto neuroscience literature showing that our frontopolar and prefrontal cortical regions naturally start waking up and working harder during late-morning REM transitions (Dresler et al., 2012). Because the exact brain regions we use for logic and self-reflection are coming back online, recognizing the impossibility of a dream world becomes much easier.

My logs also proved that my wrist band’s sensory trigger slipped into my dream plots more frequently during this morning resleep window than at any other time of night. Interestingly, while I thought daytime naps would offer the exact same receptive conditions, they never yielded the same level of successful lucid awareness. The morning resleep window remains the ultimate temporal target for this interaction.

6.2 Inside the Lucid Dreams

The autobiographical journal entries demonstrated that waking habits can successfully transfer into dream behavior through repetition and embodied practice. In all successful lucid dreams that were achieved, awareness emerged because a practiced daytime reality check failed under dream conditions, specifically involving introspective inspection of the hands and fingers. I traced it back to either the long-term mental habit of my daytime training or the direct, immediate impact of the nighttime vibration. Three successful dreams (1, 2, 3) were achieved during the active experiment, and interestingly enough, one (4) occurred spontaneously after the formal experiment had ended.

1. I walked into some big storehouse. I counted my fingers, more than 13. And some of them were small weird stumps, just a few long fingers. I dreamed. I turned to my sister. “Do you love me?” She replied in our sarcastic way: “a little.” I tested out one more thing to see, took a chair and pushed it down into the floor. It stopped halfway through the floor. Then I jumped into the air and out I flew!

2. I looked down to carefully count my fingers, and saw that I had an impossible number of fingers, so many that it almost resembled a fan. This confirmed it. I was dreaming.

The environment refactored, I was in a coastal city that looked exactly like Split, populated by large groups of young adults. I navigated the streets while executing continuous physical audits—rubbing, feeling, and counting my fingers, which stabilized at six and seven digits. I arrived at a restaurant by the water inside a stone fortress. I observed details, the physical traits of the crowd, noticing everybody's faces, and someone with an exaggerated, unusual hairstyle. It was fun, and I thought to myself: this is cool, that my head created all this and I can explore it. A group was playing a boardgame and others at a table were eating potato crisps, a bag I really looked at the design on. I accidentally dropped a bag of crisps onto the ground but took it up and ate one, noting a distinct, accurate taste feedback. I walked over to a window inside the fortress and tried to force the manifestation of a person. A dark silhouette began walking toward me from around a distant corridor corner, but the mental strain fractured the sleep state, causing the environment to dissolve.

3. I was in my hometown McDonald’s with two of my sisters. We ordered green vegetable soup with a checkered waffle-pancake, which tasted delicious. We moved to a corner table near the cash register. The restaurant was crowded, and it was dark outside. I went to buy more soup, and at that moment, an unnaturally tall guy entered the restaurant. "Look how tall he is," I remarked, pointing him out to my sisters. His head almost reached the ceiling by the register, and when he walked into the restroom, he had to bend down. He returned and sat with his group at the opposite end of our row of tables. He said something that made his entire group laugh, including my sisters, though I couldn't hear what it was. In hindsight, I understood that he had asked, "Does anyone want to grab a coffee? I brought some pastries from Örebro." I reasoned he was travelling with his basketball team, as he was so all. I wondered: if he had driven all the way from Örebro and stopped to eat here in Säffle, where was he heading? Oslo? I thought to myself, Wow, McDonald's is really the place to be if you want to meet new, interesting people in this little town.

   My sisters and I began playing cards. The tall guy walked over and asked if he could join us. "Of course," I replied. He and his friends moved over, and we pushed our tables together to play. I left to use the restroom, which turned out to be the most disgusting toilet imaginable; the smell was so foul I nearly vomited. There was yellow sludge in the corner of the floor and a filthy toilet seat, but I desperately had to pee. Afterward, I washed my hands and noticed some forgotten personal items left on top of a cabinet. I decided to spray some perfume on myself.

When I came out, I saw my sisters exiting the restaurant along with a large crowd. I went back to retrieve my jacket but couldn't find it anywhere. I wondered if one of my sisters had accidentally taken it. I only found a different leather jacket—my sister's? The tall guy was still sitting there, playing cards with a few people. At this moment, I began to realize that this entire scenario was a dream. It took a lot of conscious mental effort to calm myself down and think, Look, the jacket doesn't matter at all. This is just a dream. You still have your actual jacket in real life. After that, I was totally free. Remembering the strategies I had read about, I told myself to stay calm and avoid doing anything too drastic that might shock me awake.

I sat down next to the tall guy. I looked down at my hands; they appeared almost normal, but when I carefully counted the fingers, I noted six digits. I turned my hands over, inspecting both sides, rubbing them together, and focusing on the tactile feeling, confirming: I am absolutely inside a dream. Cool. I looked back at the group playing cards, calculating my next move: Okay, how should I approach the tall guy? Keep it calm, nothing too drastic, cause that usually destroys the lucid dream. I had a smooth card to play. I caught his attention and spoke to him with a very quiet whisper. He had to bend down close to my face to hear me. The moment he leaned in, I kissed him on the cheek. He responded instantly. A fleeting thought crossed my mind—Can people actually make out inside McDonald's?—but I dismissed it as this was only a dream, I could do anything. I wrapped my arms around and looked at my hands once more to ground myself.

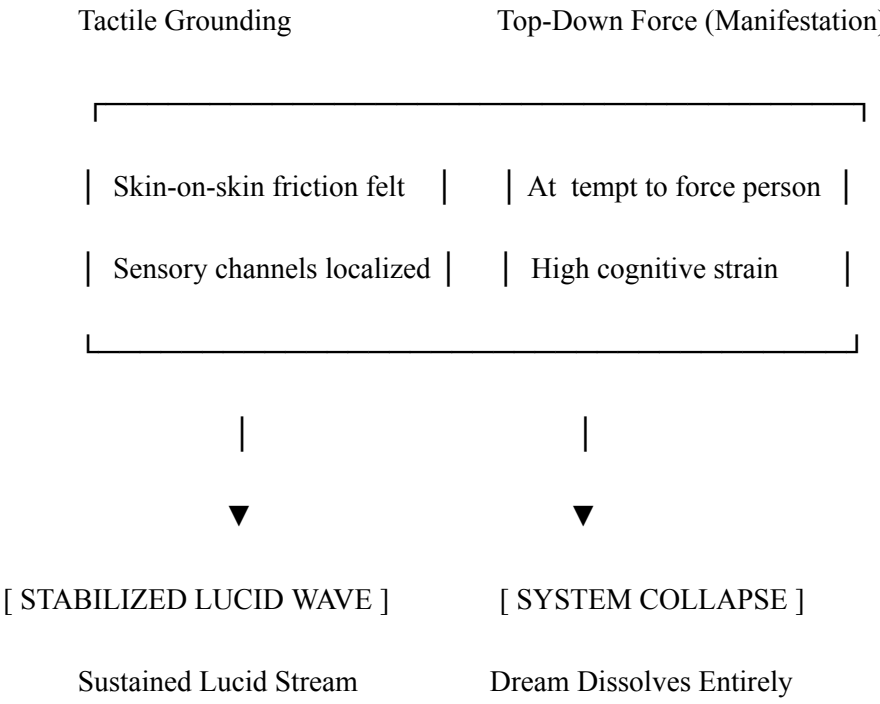
4. I took off a bunch of bracelets from my wrist. Then I looked at my wrist again, and they were all still there. This made me realize that I was in a dream.

6.3 Experiential Analysis

Once I knew I was dreaming, I was able to explore the environments with full reflective awareness and logical reasoning. However, these experiences proved that lucid dreams are incredibly fragile structures. Just like I noticed years ago in my baseline journals, any attempt to exert heavy-handed control or let emotional excitement take over immediately broke the dream, causing the landscape to dissolve into blackness or shock me awake. Keeping a dream stable depends far less on aggressively commanding the environment and much more on calm observation and maintaining a grounded physical presence.

Analyzing these successful lucid experiences gave me insights into how daytime habits and the limits of cognitive control all play together.

[MATERIAL STABILIZATION VS. NARRATIVE COLLAPSE]



Across all three dreams, my manual finger-counting routine was the primary gateway into lucidity. Looking down at my hands forced a direct, clear collision between my deeply ingrained waking body model (expecting five fingers) and the mathematically unstable rendering capacity of my sleeping brain. Interestingly, my subconscious handled this rendering failure in completely different visual ways each time:

> In Dream 1, my hand distorted into a weird mix of short, blunt stumps and elongated single fingers.
>
> In Dream 2, the mutation was much more extreme, stretching out into a dense layout that looked like a fan.
>
> In Dream 3, the glitch was remarkably subtle; my hands looked completely normal at a glance until a focused, deliberate count revealed an extra sixth digit.

This progression proves an interesting design point: while visual details in dreams fluctuate wildly, our tactile and proprioceptive awareness of our own body remains incredibly resilient. In Dream 3, the moment I spotted the visual glitch, I instinctively stabilized the entire dream by turning my hands over

and rubbing them together. Focusing intensely on the physical feeling of skin-on-skin friction acted as a somatic anchor. This physical grounding slowed down my emotional intensity, letting me transition from a passive spectator into an active participant without accidentally shocking myself awake.

The log entries highlight a strict operational threshold when it comes to having agency inside a dream: creative intention works beautifully if you maintain a detached, low-attention mindset, but it completely crashes under high-strain, top-down command.

In Dream 3, I managed to pull off a sophisticated level of narrative control simply by practicing emotional restraint. The moment I realized my jacket was gone, I bypassed the usual dream panic by telling myself: *"Look, the jacket doesn't matter at all. This is just a dream."* I consciously suppressed my excitement and kept my actions incredibly subtle because I knew that overreacting destroys lucidity.

Dream 2 shows exactly what happens when you push too hard. When I tried to aggressively force a specific person to manifest around the corner, my brain struggled to compute the heavy, top-down demand. The dream engine could only render a faint, unstable dark silhouette before the sheer mental strain fractured my sleep state, causing the entire environment to instantly melt away into waking reality.

This confirms a foundational rule: interactive devices must never encourage a user to forcefully dominate, micro-manage, or fight their dream environments. Instead, the technology needs to work like temporary physical scaffolding—simply helping the user maintain a state of calm, mindful, and receptive bodily presence.

6.3.1 Autonomous Conditioning

An unexpected finding emerged right after I wrapped up the active tracking phase of the experiment. During a post-experimental monitoring period—where I completely took off the physical wearable and stopped running any daytime or nighttime triggers—I experienced a spontaneous lucid dream.The wristband theme obviously leaked into my dreamscape because my brain had adapted to the prolonged tactile pressure of the band I wore throughout the weeks of the experiment.

Within interaction design and behavioral psychology, this cognitive bleed-through changes how we think about wearable architectures. It proves that the *Metacognitive Bridge* is more than a localized technological crutch. Instead the repeating daily association between a specific body part (the wrist), a distinct material presence, and an intentional mental disruption became deeply woven into my internal cognitive architecture. Over time the biological brain actually learns to build a simulated version of the interactive device, letting the habit replicate itself autonomously.

This long-term residual effect shows that somaesthetic interfaces can work as genuine tools for internal education. By anchoring technology to consistent bodily locations and mindful daytime rituals, we can look past short-term data collection. Designers can establish enduring, cross-state habits that continue to reshape a user's personal awareness long after the physical hardware is sitting in a drawer.

Traditional consumer electronics build loops of system dependency; if you take off your fitness tracker or smartwatch, your behavioral motivation usually collapses. In this project, the design process serves a radically different purpose: it works as a temporary scaffolding mechanism. We deploy the physical artifact to consciously train the nervous system, but its ultimate success is achieved when the technology can be removed, leaving behind a permanently heightened baseline of autonomous human introspection.

6.3.2 Synthesis of Autobiographical Experiment

While the field study mapped out the qualitative human values of the project, my 24-hour reality loop self-experiment pinned down the hard physiological, sensory, and chronobiological parameters needed to actually make the system work. Living through this rigorous self-testing protocol stripped away my speculative assumptions by slamming the design requirements right into the concrete material limitations of the human body during sleep.

The active findings demonstrated just how critical sensory continuity is across our waking and dreaming states. Repeating my daytime exposure to an identical, highly specific tactile signal directly trained my brain to recognize that exact same stimulus when it fired mid-dream. This proves that reflective dream awareness thrives on gradual habit formation and physical repetition, rather than on sudden, jarring, or highly intrusive shocks to the nervous system. Keeping the sensory signature and its anatomical location strictly identical creates a gentle cognitive ramp, helping the mind transition safely from passive dreaming into active self-reflection.

[Daytime Somatic Pulse] —> Deeply Internalized Habit —> [Identical Nighttime Cue Recognized]

The experiment also proved that the fine-grained sensory qualities of the trigger dictate its entire success. Generic smartphone vibrations were a total failure—they were constantly misidentified, ignored, or completely swallowed by passive dream logic. It takes a hybrid cue, blending a unique tactile rhythm with a subtle audio chime, to slice through the noise and create a distinct perceptual anchor. Similarly, where I put the hardware made or broke my sleep continuity. Shoving the device inside my pillowcase jounced me awake in a panic, whereas wearing it on my wrist delivered a much softer, less disruptive pulse that let the cue blend naturally into the dreamscape.

The numbers established clear temporal boundaries for when these cues should actually fire. Running continuous, night-long tracking or blasting indiscriminate cues was an absolute nightmare, causing severe sleep fragmentation during my early-night tests. Instead, lucid awareness and sensory incorporation occurred almost exclusively during my late-morning “resleep” periods following a brief awakening. This pattern aligns perfectly with neuroscience literature showing that our reflective cognitive processes naturally come back online during late-morning REM transitions (Dresler et al., 2012). The lesson here is clear: the interaction must target these highly receptive, transitional morning windows instead of trying to micro-manage the entire night.

The spontaneous, autonomous conditioning I experienced after taking the hardware off shows that pairing a fixed body location (the wrist) with a randomized daytime prompt can permanently rewrite your cognitive schema. This lingering residual effect suggests that the physical artifact functions as temporary behavioral scaffolding. We are not designing a permanent technological crutch; we are building a tool to train the nervous system, ultimately empowering the mind to execute deep self-reflection completely independent of active hardware triggers.

6.4 The Architectural Rationale

A central design question emerging from these combined syntheses concerns why the interaction requires a dedicated physical artifact rather than a smartphone application alone. Although smartphones contain advanced haptic systems and computational capabilities, they remain behaviorally associated with wakeful productivity, task-switching, social communication, and digital anxiety. Their presence within the sleep environment introduces physiological contradictions that conflict directly with healthy sleep conditions.

Smartphone interactions are fundamentally screen-based and visually stimulating. Blue-light exposure and notification-oriented interaction patterns disrupt melatonin regulation and increase cognitive activation before sleep (Walker, 2017). Using a glowing touchscreen interface to configure or calibrate a dream-priming system right before bed actively disrupts the user's autonomic nervous system, conflicting with the soft, vulnerable conditions required for rest. The *Metacognitive Bridge* requires a completely screenless, analog ritual to lower sensory anxiety and protect the sleep sanctuary from digital clutter.

A smartphone notification is semiotically engineered around the physics of wakeful disruption. Mobile phone vibrations are universally associated with task-switching anxieties, such as text messages, incoming emails, and social media reminders. When a smartphone buzzes near a sleeper's head, the brain immediately parses the signal through this wakeful lens. This leads either to an abrupt, startled awakening or a total misidentification of the cue within the dream narrative.

A phone loosely placed in a bed changes physical orientation constantly throughout the night. The final wristband concept avoids this digital fatigue by enforcing an absolute Somatic Coherence. By remaining strapped to the exact same anatomical position (the wrist) continuously across the 24-hour cycle, the trigger transforms from an intrusive, external notification toward functioning as a familiar bodily signal integrated into everyday experience.

A smartphone is structurally incapable of maintaining continuous, skin-safe biometric contact with the body. To transition dream engineering away from speculative timers toward precise, automated delivery, the system would in future work utilize real-time Photoplethysmography (PPG) sensors embedded directly against the epidermis of the wrist. By continuously measuring the user's pulse and heart rate variability, the wearable's internal software loop maps the precise drop in autonomic activity that characterizes deep sleep, as well as the sudden heart-rate irregularities that signal a transition into a stable Rapid Eye Movement (REM) wave. Decoupling this computational loop from a phone would allow the artifact to operate as a self-contained, automated tool for individual metacognitive habit formation.

6.5 Design Requirements

The combined findings from the field study and my autobiographical experiment generate several concrete design requirements for the final artifact:

> Triggers placed near the head are structurally unsuited for gentle sleep interaction, as they consistently cause abrupt, startled awakenings and ruin sleep continuity. Consequently, the hardware is optimized as a wristband. This maintains stable, continuous contact with the body across both halves of the day, creating a seamless somatic link between separate states of consciousness.
>
> To support prolonged use across a continuous 24-hour cycle, the wearable requires a lightweight, flexible, and soft material housing profile. Comfort is essential here—the device needs to minimize physical distraction during wakeful productivity while avoiding painful pressure points or catching on fabric while tossing and turning in bed.
>
> Standard smartphone vibration motors produce generic, diffuse sensations that are easily misidentified. Combining unique tactile rhythms with subtle acoustic tones creates a much stronger sensory identity. This creates a non-associative somatic cue that the mind can easily distinguish from standard communication-based smartphone alerts.
>
> Fiddling with illuminated smartphone interfaces or touchscreens around sleep introduces harsh visual stimulation and disrupts essential sleep hygiene. The system should therefore

support fully screenless interaction through physical analog controls and low-attention calibration methods to preserve a quiet, screen-free bedroom ritual.

Because the highest probability of sensory incorporation and reflective awareness occurs during transitional REM waves (the "resleep" window), the user interface needs to cater directly to this state.

7. Design Process & Prototyping

7.1 Form Generation

To put the principles of somaesthetic interaction design (Höök, 2018) into practice, I intentionally steered the physical form of the wearable device away from the rigid, glossy plastics typical of commercial fitness trackers. Instead my initial ideation focused on creating an interface that feels like a comforting, familiar textile rather than a cold, clinical monitoring tool. Because this device is designed for the intimate context of sleep, its material qualities directly shape the user's sensory experience as a wearer.

This tactile focus responds directly to the empirical findings from my field study and autobiographical experiment. The final form minimizes both wakeful cognitive friction and nighttime physical obstruction to support continuous body placement across a full 24-hour cycle.

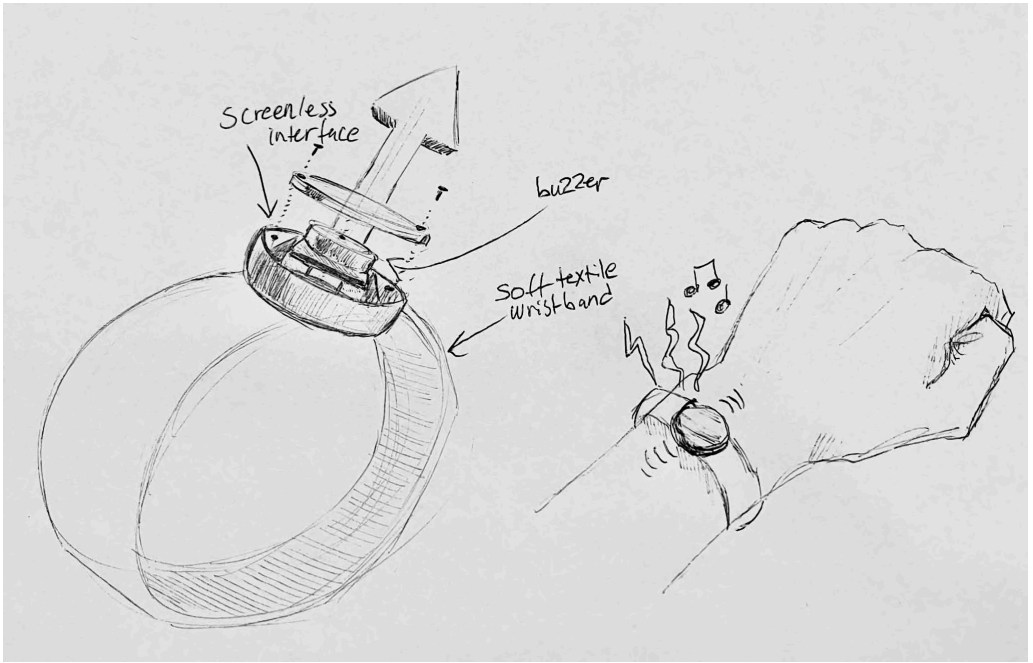


Figure 3: Hardware sketch, exploded view and in use.

Within interaction design, Color, Material, and Finish (CMF) parameters serve as primary tools for guiding a user's emotional and behavioral responses to an object (Karana et al., 2014):

> Color: Specific color wavelengths can significantly alter our psychology. Warm tones like red or yellow tend to stimulate the nervous system, whereas cool, desaturated colors—particularly blues and greens—induce a sense of quiet and cognitive comfort (Elliot & Maier, 2014).
>
> Material: The structural characteristics of a fiber determine its physical and emotional utility. Plant-based options like cotton offer a light, structured, and breathable layer, while organic animal fibers like wool provide thermal warmth and a deep sense of physical security.
>
> Finish: The surface texture defines an object's tactile identity. Soft finishes lower sensory anxiety, reduce negative feelings, and help counter the synthetic, "cold" feeling often associated with digital electronics (Shusterman, 2008).

I chose a muted, desaturated sage green for the wristband to bring an immediate sense of calm and environmental grounding to the user. In design, green is strongly tied to rest and balance (Elliot & Maier, 2014). Studies show that exposure to natural tones can signal psychological safety, helping to lower stress hormones and calm the autonomic nervous system before sleep (Ulrich, 1983). Rather than being a purely aesthetic choice, the color supports a state of relaxed alertness, a psychological balance often used in therapeutic environments to prevent mental fatigue (Kaplan, 1995).

Color psychology indicates that a shade's impact changes drastically depending on its brightness and saturation. While dark emerald greens can feel overly corporate or luxurious, and bright mint greens carry clinical, medical connotations, a desaturated sage green functions as an earthy, calming element (Akers et al., 2012). It captures the stress-relieving qualities of a natural environment without overstimulating the user.

The strap itself is hand-crocheted from a soft organic textile to meet the demands of 24-hour wear. Aligning with Shusterman’s (2008) focus on bodily mindfulness, the textile looks and feels lightweight, acting and shaping itself almost like a "dream cloud." This soft format deliberately cushions the boundary between the hard electronics (the Arduino board and piezo buzzer) and the human wrist. Nesting the technical components safely inside the forgiving textile sleeve solves a practical ergonomic issue: it keeps the rigid hardware from rubbing against the skin. This combination balances the user's emotional and physical needs, optimizing comfort during long self-testing windows and ensuring that the technology always remains secondary to the lived experience of bodily reflection (Corkhill et al., 2014).

7.2 Hardware Architecture and Interactive Simulation

To evaluate the actual, physical qualities of *The Metacognitive Bridge* on the human body, I built a high-fidelity interactive prototype. The hardware architecture relies on an Arduino Nano 33 IoT microcontroller mounted to a breadboard, wired to a physical push-button, and paired with a low-profile piezo buzzer.

In the envisioned commercial version, the device would operate autonomously on a true 24-hour cycle, delivering unpredictable triggers separated by hours of complete silence to prompt real-time waking mindfulness and nocturnal anchors. However, during a user study or design examination, testing such massive time intervals is logistically impossible.

To solve this, I designed the prototype to compress this expansive temporal framework into an accelerated interactive simulation. By pressing the physical button, the user kicks off a localized, rapid cycle: the system pauses in total silence for a randomized window of 2 to 10 seconds before automatically firing a distinct, recognizable acoustic melody. This compression allows users to instantly experience and test the sudden transition from baseline focus to sensory-induced metacognitive auditing right on their arm, making a long-term psychological process immediately tangible during design critiques.

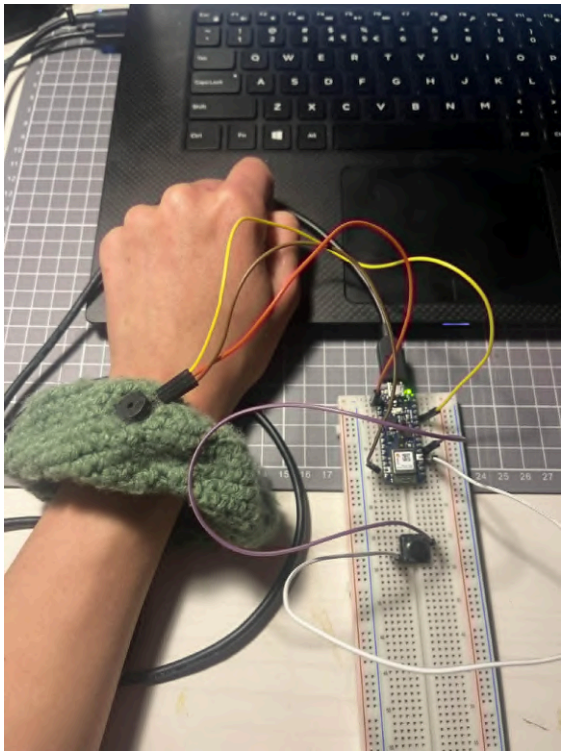


Figure 4: The interactive hardware prototype.

7.3 Code Implementation

The underlying software architecture was written in C++ to craft the emotional and aesthetic qualities of the acoustic signal. Through iterative user testing with various melodies, tempos, and sound structures, the code was optimized to play an uplifting, recognizable motif in a bright C-Major scale. The notes are programmed to glide into one another at an accelerated tempo to provoke an alerting shift in attention.

C++

```
int buzzerPin = 8;
int buttonPin = 2;

// High-frequency, bright acoustic tones (C-Major Scale)
int noteC4 = 262;
int noteE4 = 330;
int noteG4 = 392;
int noteA4 = 440;
int noteC5 = 523;

void setup() {
  pinMode(buzzerPin, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP); // Utilizing internal pull-up resistor for button stability

  // Seeding the random generator using open-pin background analog noise ("noise")
  randomSeed(analogRead(A0));
}

void loop() {
  // Check if the user initiates the interactive simulation loop
  if (digitalRead(buttonPin) == LOW) {

    // Calculate a randomized delay between 2 and 10 seconds
    long randomSeconds = random(2, 10);
    long waitTime = randomSeconds * 1000;

    // Pause and maintain complete sensory silence
    delay(waitTime);

    // -- PLAY THE UPLIFTING MELODY --
    // Notes are executed quickly to prompt alertness,
    // blending smoothly together for an uplifting feeling.

    tone(buzzerPin, noteC4);
    delay(200); // Swift, energetic start

    tone(buzzerPin, noteE4);
    delay(200);

    tone(buzzerPin, noteG4);
    delay(200);

    tone(buzzerPin, noteA4);
```

```
    delay(200);

    // Triumphant resolution tone sustained slightly longer for memory anchoring
    tone(buzzerPin, noteC5);
    delay(600);

    // Terminate acoustic feedback (Silence / Automated Reset)
    noTone(buzzerPin);

    // Interaction Guard: Halts code progression until the button is fully released
    while (digitalRead(buttonPin) == LOW) {
      delay(10);
    }

    delay(100); // Debounce interval for system stability
  }
}
```

7.4 Code Logic and Architectural Breakdown

During setup(), the system reads electromagnetic static noise from an empty analog pin (A0) to create a genuinely unpredictable randomization curve.

When the input loop registers a low signal from the push-button (INPUT_PULLUP), it calculates a randomized variable between 2,000 and 10,000 milliseconds. The code executes a delay(waitTime), forcing the system into absolute silence to simulate the "low-attention" state of daily life or calm sleep.

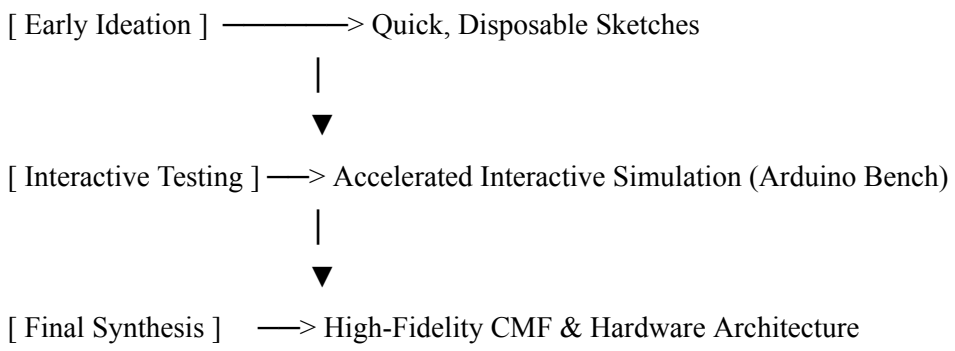
Once the delay expires, the code steps rapidly through five distinct pitches, building up to a sustained noteC5. This progression creates an intentional "triumphant" finish designed to stand out against environmental background noises without triggering anxiety.

A locking while loop halts code progression as long as the button remains pressed. This ensures that a single physical interaction yields exactly one simulated reminder loop, resetting the system for subsequent user tests.

7.5 Conceptual Prototyping

To ground the trajectory of my physical design process, it helps to look at the prototype's evolution through Bill Buxton's (2007) framework of Design Sketching. Buxton draws a clear line between a sketch—which is quick, cheap, disposable, and meant to suggest an idea—and a prototype, which actively tests, refines, and answers concrete engineering questions.

BUXTON'S DESIGN FUNNEL



The hand-crocheted sage green sleeve functioned as an experience sketch in this project. It was never meant to operate as a high-fidelity diagnostic instrument. Instead, it was a cheap, tactile probe built to map the body's physical and emotional compatibility with an interface.

By scaling down the time framework from a full 24-hour cycle to an accelerated simulation loop (2 to 10 seconds), this material sketch let me instantly test the sudden shift from baseline focus to sensory-induced metacognitive auditing on my arm. This progression reflects the classic design funnel, moving systematically from loose, ambiguous material sketches to the rigid, high-fidelity microcontroller architecture needed for data collection.

8. Knowledge Contributions

A primary objective of the Research through Design (RtD) methodology is to generate transferable, abstract design knowledge by deliberately materializing and critiquing physical artifacts (Zimmerman et al., 2010). For my project, developing and empirically testing *The Metacognitive Bridge* did not merely solve a localized technical problem; instead, it yielded programmatic insights into how interactive systems interface with marginal human cognitive states. By decoupling technology from conventional, screen-based notification paradigms and anchoring it in intermediate bodily rhythms, my thesis presents a distinct conceptual contribution to the field of Interaction Design.

CROSS-STATE KNOWLEDGE CONTRIBUTION



Proposed Cross-State Paradigm:

[Wakeful Priming] ──(Somatic Consistency)──> [Anchor] ──(Recall Optimization)──> [Waking Introspection]

8.1 Designing for Low-Attention States

Interaction design has historically operated under an industry-wide bias toward wakeful productivity, task optimization, and highly active cognitive engagement. When conventional frameworks address low-attention scenarios, they typically treat them as temporary operational deficits to be bypassed or corrected. This work contributes a counter-framework by positioning the human subconscious not as an operational void, but as an active, highly receptive site for somatic interaction.

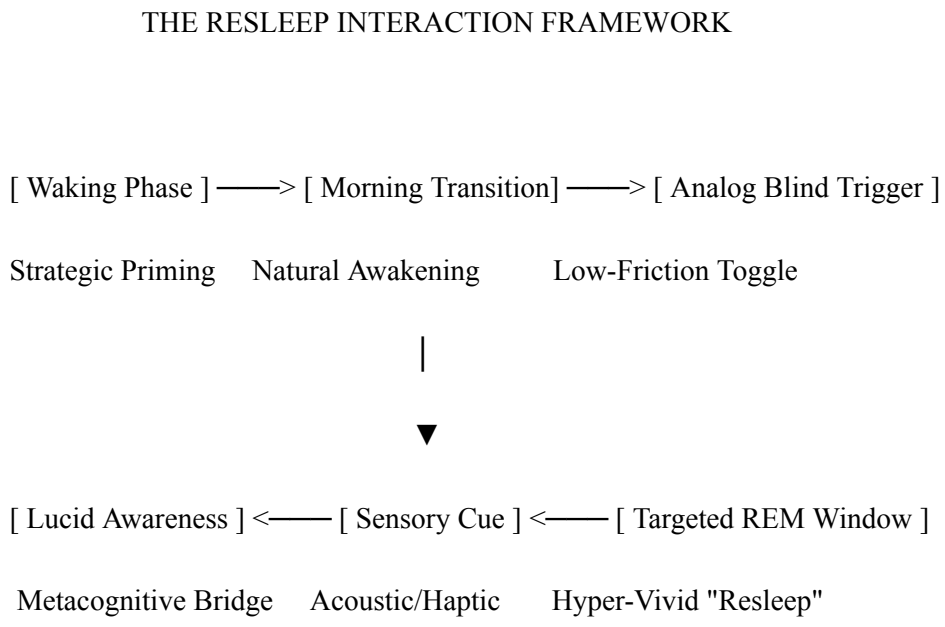
Designing for the subconscious requires a profound shift away from the semantics of information display toward the semantics of somatic resonance. When a user is drifting through sleep architecture, the mind lacks the top-down executive logic centers, specifically the deactivated dorsolateral prefrontal cortex (Muzur et al., 2002), necessary to parse semiotic, abstract data or complex visual hierarchies. Therefore we cannot communicate via symbolic indicators like digital text alerts or glowing application interfaces. Instead data must be translated into non-associative, multi-sensory somatic anchors that map directly to the primitive sensory processing pathways that remain online during sleep (Scarpelli et al., 2021).

By demonstrating that a hybrid acoustic-tactile trigger can interact with dream architecture without fracturing sleep continuity, this research provides a baseline lexicon for subconscious interaction design. It suggests that the material qualities of an interface, its rhythm, physical location on the body, and structural flexibility, are the primary language through which technology can converse with altered states of human consciousness.

8.2 The Resleep Interaction Framework

Traditional commercial sleep engineering relies on a paradigm of continuous, night-long surveillance, tracking micro-fluctuations in heart rate variability and blood oxygen saturation to map static sleep data retrospectively. In contrast, this thesis introduces the *Resleep Interaction Framework*, which isolates specific high-receptivity temporal windows within human sleep architecture to compress, target, and optimize sensory interactions.

The framework standardizes the transitional sleep boundary (specifically the brief period of awakening followed by an immediate return to sleep) as a highly fertile, low-friction playground for interaction design prototyping. Rather than attempting to maintain strict signaling across an unpredictable eight-hour sleep cycle, the *Resleep Framework* focuses exclusively on hyper-vivid morning REM waves, where the brain's executive logic centers are naturally beginning to come back online (Dresler et al., 2012).



To operationalize this window, the framework dictates a specific two-part interactive sequence:

The device architecture must provide an analog control interface that can be executed flawlessly on the sleep boundary without visual attention or cognitive friction. This completely bypasses the blue-light disruption and notification anxiety induced by modern smartphone dashboards.

Upon activation, the system automatically schedules a compressed, highly targeted sequence of non-associative cues tailored to fire precisely within the subsequent, predictable REM wave.

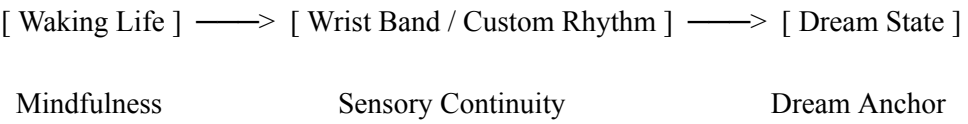
By treating sleep as a dynamic, fluctuating landscape with specific design-accessible entry points, the *Resleep Interaction Framework* provides future researchers with a systematic method to inject sensory prompts where the probability of successful cognitive incorporation and metacognitive reflection is highest.

8.3 Somatic Anchors vs. Digital Alerts

This thesis formalizes a distinct conceptual dichotomy within wearable product design: the structural divide between an attention-grabbing digital alert and a continuous somatic anchor.

Modern commercial wearables are engineered around the physics of disruption. They utilize standard eccentric rotating mass (ERM) vibration motors to deploy generic, diffuse tactile buzzes. Because these mechanical patterns are heavily associated with wakeful, high-anxiety communication networks, such as work emails, text messages, calendar reminders, and push notifications, they carry an immense cognitive load and are immediately categorized by the mind as external disruptions. When these generic patterns bleed into sleep, the brain either completely misidentifies them as real-world threats or shatters sleep architecture to trigger an abrupt, startled awakening.

A somatic anchor, by contrast, establishes an unbroken, 24-hour baseline of bodily familiarity. It achieves this by enforcing a strict sensory and material consistency across separate domains of consciousness.

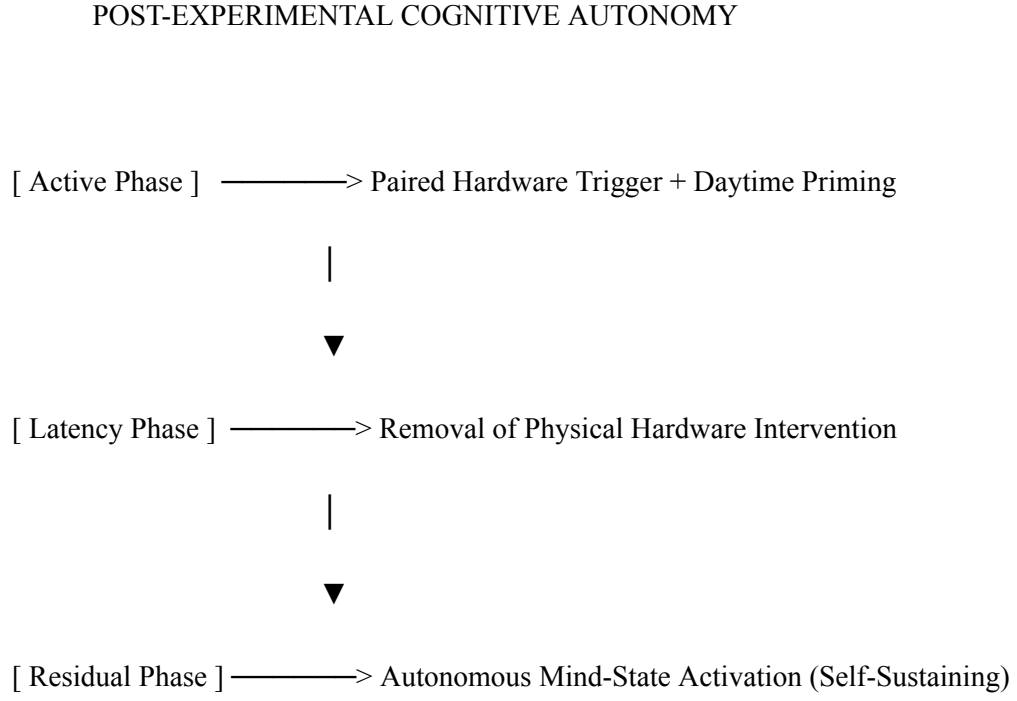


By keeping the physical body location (the wrist), the material interface constraints, and the custom rhythmic texture of the hybrid trigger completely identical between day and night, *The Metacognitive Bridge* creates a specialized sensory vocabulary. The trigger does not pull you out of your environment to deliver an abstract data update; instead, it enters your environment as a familiar somatic baseline.

Practicing this mindful physical connection during the waking day primes the nervous system. When the identical cue fires during the night, the brain identifies it not as an intrusive real-world alarm, but as an internal catalyst for reflective awareness. This design shift demonstrates how wearables can move away from aggressive data extraction, transforming themselves into calm, comforting material extensions of human bodily consciousness.

8.4 Post-Artifact Autonomous Conditioning

A significant, unexpected knowledge contribution emerged from the long-term cognitive residuals observed during the post-experimental monitoring phase. The occurrence of an autonomous lucid instance following the removal of the physical prototype suggests a phenomenon I term *Post-Artifact Autonomous Conditioning*.



In traditional Human-Computer Interaction (HCI), design models are typically optimized for system retention and engagement, inadvertently creating an interactive loop of permanent user dependency. This research illustrates an alternative paradigm: interaction design can function as a temporary behavioral scaffolding tool. By coupling a fixed anatomical position with an unpredictable wakeful habit over an extended period, the biological brain appears capable of internalizing the interaction rules, eventually recreating the system logic entirely within the subconscious dreamscape without active hardware triggers.

The ultimate success of a somaesthetic interface may therefore be measured by the point at which the physical technology can be removed, leaving behind a heightened baseline of human introspection and cognitive agency. This long-term residual effect supports the possibility that somaesthetic interfaces can act as genuine tools for internal education. By anchoring technology to consistent bodily locations and mindful daytime rituals, designers can move past short-term data collection,

establishing enduring cross-state habits that continue to reshape personal awareness long after the physical hardware is taken off.

9. Critical Reflection & Conclusion

9.1 Methodological Boundaries and Subjectivity

Using an autobiographical design methodology was the only way I could capture the fleeting, fragile nuances of transitional sleep states. A detached user study simply cannot access these moments reliably. Pinning down the exact feel of a tactile rhythm, evaluating acoustic volumes under low-attention conditions, and tracking how waking habits bleed into dream mechanics required me to stay in an intimate, continuous loop of self-observation and rapid prototyping. An essential part of Research through Design (RtD) rigor means acknowledging that these highly personal insights come with obvious subjective boundaries.

Human sleep architecture, stress levels, baseline dream recall, and skin sensitivity vary wildly from person to person (Carskadon & Dement, 2011). To maintain academic validity, I must clarify that the findings from my "24-Hour Reality Loop" experiment point toward localized interaction possibilities rather than universal human benchmarks (Neustaedter & Sengers, 2012).

To keep this inquiry completely transparent, I treat my material failures with the exact same analytical weight as my three successful lucid instances. Documenting the sleep fragmentation, like shoving the device into my pillowcase, or the moments where haptic buzzes were completely swallowed by the sleep, ensures this study stands as an honest piece of design research rather than a forced narrative of perfect technical success.

Designing interactions for the intimate sanctuary of sleep requires constant ethical questioning. Modern consumer culture is obsessed with productivity metrics, routinely framing sleep as a mere biological optimization window for post-industrial recovery or physical data surveillance. Blasting active, hardware-triggered cues into this private, restorative state carries real risks of sleep fragmentation, micro-arousals, and unwanted cognitive friction.

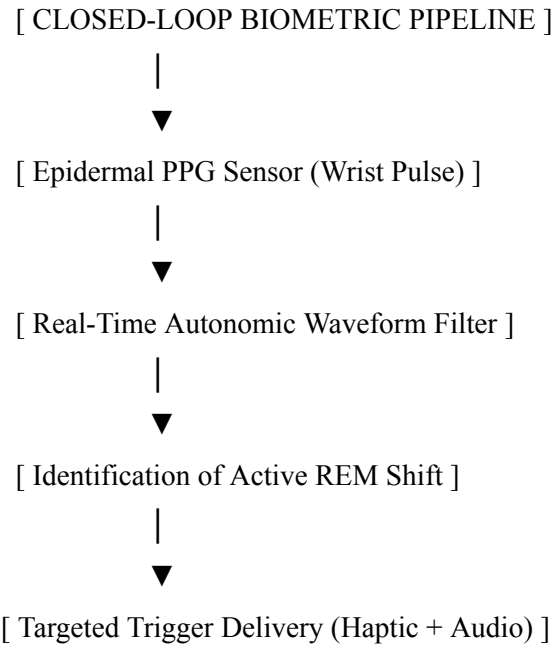
I intended *The Metacognitive Bridge* to serve as a critical alternative to data-heavy consumer tracking. The artifact deliberately avoids cloud-connected infrastructure, blue-light-emitting screens, or algorithmic sleep-efficiency scores. It prioritizes simplicity and manual, analog controls, ensuring that technology stops short of intrusive surveillance at the bedroom threshold.

My design does not try to force behavioral compliance or turn sleep into an arena of measurable performance. Instead it works to provide a calm material context for personal reflection, keeping the technology positioned purely as a supportive tool for bodily self-awareness.

9.2 Future Work

To take *The Metacognitive Bridge* beyond its exploratory autobiographical boundaries, my next step is to scale the current design requirements into a deployable batch of multi-user testing kits. The immediate mechanical task involves refining the 3D-printed enclosure geometries and electronics housing to improve durability, sweat resistance, and skin-safe component encapsulation for extended home use.

A core objective for future versions is moving away from the prototype's semi-speculative temporal alarms toward an automated, bio-adaptive closed-loop delivery architecture. Optimizing the nocturnal anchor requires delivering the sensory cue precisely during active Rapid Eye Movement (REM) sleep.



To achieve this, future design iterations will integrate a low-profile photoplethysmography (PPG) pulse meter flush against the underside of the wristband, keeping continuous contact with the skin over the radial artery. The localized software loop will continuously monitor the user's resting pulse and heart rate variability. By algorithmically identifying the characteristic drop in sympathetic nervous system tone and the irregular pulse fluctuations that signal entry into a stable REM wave, the system can self-trigger the hybrid haptic-acoustic cue at the exact physiological moment the sleeping mind is most receptive to it.

Following this technical optimization, expanded field studies will distribute these wearables to a diverse participant group over a multi-week tracking period. This expansion will shift my evaluation parameters away from individual observation toward a broader collective critique. Long-term user testing will reveal how different bodies, domestic routines, and unique sleep environments handle automated biometric cueing, proving whether these bio-adaptive loops can reliably foster dream accessibility without ruining rest quality.

9.3 Conclusion

This thesis investigated how an interaction design system can support dream accessibility by developing intentional, cross-state somatic habits. While the visual artifacts from *The Dreamer's Pillowcase* cultural probe exposed a widespread, collective recall gap—a clear tension where users voice a strong desire for dream agency but lack any meaningful interaction tools—my autobiographical experiment demonstrated that a structured 24-hour reality loop can build an operational bridge between separate domains of human consciousness.

By mapping an abstract daytime mindfulness practice onto a distinct, worn sensory trigger, the system successfully transferred waking cognitive conditioning across the threshold of sleep. When these physical, acoustic-tactile prompts glitched under the altered laws of dream physics (manifesting inside my dreams as multiplying fingers, conversational logic breakdowns, or structural material anomalies) they transformed the passive dream landscape into an accessible site for active self-reflection.

This project shows that interaction design can meaningfully engage with altered human states without resorting to mystical framing or ungrounded claims. Technology’s relationship with the sleeping body can evolve beyond the passive tracking of cold graphs and data extraction. By prioritizing analog rituals, interaction design can transform sleep from an isolated period of biological recovery into a meaningful, embodied extension of the reflective self, connecting both halves of our lives.

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Appendix A: 24-Hour Reality Loop Design Journal Template

Part 1: The Evening Log (Pre-Sleep)

Document the waking phase and set your intentional "somaesthetic" anchor.

General Info

- Date: _____ | Bedtime: _____
- Waking Mood/Energy: _____
- Daily Activity/Stressors: _____
- Screen time: _____

Metacognitive Priming

- Daytime Triggers and Times: _____
- Reality Checks Performed: [ ] Digital/Text [ ] Solid-Push [ ] Finger Counting [ ] Other: _____
- Somaesthetic Reflection: *How did the trigger feel today? (e.g., Sharp, fun, annoying?)* _____

Nighttime Sensory Setup

- Modality: [ ] Vibration Only [ ] Sound Only [ ] Hybrid (Both)
- Pattern/Tone Profile: *(e.g., 3 short pulses, continuous low hum, rising pitch)* _____
- Device Placement: [ ] Wrist [ ] Inside Pillow (by head) [ ] Nightstand [ ] Other: _____
- Rationale: *Why this specific setup tonight? (e.g., Testing if sound is less disruptive)* _____

Part 2: The Morning Log (Post-Wake)

Complete IMMEDIATELY upon waking to capture subtle sensory nuances.

General Sleep Info

- Wake-up Time: _____
- Sleep Quality (1–5): _____ | How do you feel now? *(Groggy, alert, emotional?)* _____
- Number of Awakenings: _____

The Nocturnal Anchor (Somaesthetic Analysis)

- Trigger perceived during sleep? [ ] Yes [ ] No
- Sensory Texture: *Describe the feeling of the trigger.* _____

- Trigger Time: _____
- Interaction Type:
    - [ ] Full Awakening: *It pulled me out of sleep.*
    - [ ] Narrative Incorporation: *It entered the dream.*
- Achieved Lucidity? [ ] Yes [ ] No
    - [ ] DILD (Dream-Induced): *You were dreaming, felt the trigger, and then became lucid.*
    - [ ] MILD (Mnemonic-Induced): *You became lucid because you remembered your daytime training, independent of the trigger.*
- How the lucidity was achieved in the dream:

  _____
- Comments:

  _____

Dream Documentation

- Dream Summary:

  _____
- Theme & Feeling:

  _____
- Body & Feeling:

  _____

Appendix B: Malmö University AI Use Disclosure Statement

This statement outlines how Large Language Model (LLM) platforms were used and integrated throughout this Master's thesis project, satisfying the compliance metrics of the Malmö University Guidelines on AI Tools in Education (VT26).

Scope of Applied Platforms

OpenAI ChatGPT (GPT-4o Architecture) and Google Gemini (Advanced/1.5 Pro).

Formatting verification, code optimization for C++ microcontroller loops, and citation double-checking under APA 7th edition rules.

Across the course of this project I used generative tools strictly to handle text polishing and structural cleanup. The core conceptual definitions, Research through Design (RtD) frameworks, field-study, and autobiographical experiment findings were derived entirely from my own subjective design practice.

The specific interactive features and concepts were entirely directed and contextualized by me as the researcher.

The software logic provided in Chapter 7 was written by me and cross-checked through the compiler interface to verify execution stability using the generative tool as an accelerated troubleshooting bench.

In strict compliance with Malmö University's academic integrity regulations I maintain full responsibility and ownership for all arguments, literature reviews, hardware synthesis conclusions, and analytical interpretations presented within this thesis. All external quotes, conceptual schemas, and design examples have been appropriately attributed using standard academic citations within the main text and organized within the alphabetized references. The final draft has been processed through the original plagiarism prevention engine prior to university repository archiving.